

# PERHITUNGAN JARAK GEOGRAFIS MENGUNAKAN FORMULA HAVERSINE

Sistem Rekomendasi Paket Wisata Berbasis FCM

## 1. APA ITU FORMULA HAVERSINE?

Formula Haversine adalah rumus matematika yang digunakan untuk menghitung **jarak terpendek** antara dua titik di permukaan bumi berdasarkan koordinat **Latitude** (lintang) dan **Longitude** (bujur). Formula ini memperhitungkan bahwa bumi berbentuk bola, sehingga jarak yang dihitung adalah jarak di sepanjang permukaan bumi (*great circle distance*), bukan garis lurus.

## 2. FORMULA MATEMATIKA

Formula Lengkap:

$$a = \sin^2(\Delta\phi/2) + \cos(\phi_1) \times \cos(\phi_2) \times \sin^2(\Delta\lambda/2)$$

$$c = 2 \times \text{atan2}(\sqrt{a}, \sqrt{1-a})$$

$$d = R \times c$$

| Simbol                 | Keterangan                                      | Nilai/Satuan            |
|------------------------|-------------------------------------------------|-------------------------|
| $\phi_1, \phi_2$       | Latitude titik 1 dan 2                          | Derajat (°) atau Radian |
| $\lambda_1, \lambda_2$ | Longitude titik 1 dan 2                         | Derajat (°) atau Radian |
| $\Delta\phi$           | Perbedaan latitude ( $\phi_1 - \phi_2$ )        | Radian                  |
| $\Delta\lambda$        | Perbedaan longitude ( $\lambda_1 - \lambda_2$ ) | Radian                  |
| R                      | Radius bumi                                     | 6371 km                 |
| d                      | Jarak hasil perhitungan                         | Kilometer (km)          |

### 3. CONTOH PERHITUNGAN KONKRET

#### Data Koordinat:

| Lokasi          | Latitude | Longitude |
|-----------------|----------|-----------|
| Hotel di Malang | -7.9445° | 112.6178° |
| Wisata Batu     | -7.9626° | 112.6365° |

#### Langkah Perhitungan:

##### Step 1: Hitung Perbedaan Koordinat

$$\Delta\text{lat} = \text{lat}_{\text{Batu}} - \text{lat}_{\text{Malang}} = -7.9626 - (-7.9445) = -0.0181^\circ$$
$$\Delta\text{lon} = \text{lon}_{\text{Batu}} - \text{lon}_{\text{Malang}} = 112.6365 - 112.6178 = 0.0187^\circ$$

##### Step 2: Konversi Derajat ke Radian

$$\text{Rumus: radian} = \text{derajat} \times \pi / 180$$

$$\Delta\text{lat (rad)} = -0.0181 \times \pi / 180 = -0.000316 \text{ rad}$$
$$\Delta\text{lon (rad)} = 0.0187 \times \pi / 180 = 0.000326 \text{ rad}$$

$$\text{lat}_{\text{Malang}} \text{ (rad)} = -7.9445 \times \pi / 180 = -0.138659 \text{ rad}$$
$$\text{lat}_{\text{Batu}} \text{ (rad)} = -7.9626 \times \pi / 180 = -0.138975 \text{ rad}$$

##### Step 3: Hitung Komponen 'a'

$$a = \sin^2(\Delta\text{lat}/2) + \cos(\text{lat}_{\text{Malang}}) \times \cos(\text{lat}_{\text{Batu}}) \times \sin^2(\Delta\text{lon}/2)$$

$$\sin^2(\Delta\text{lat}/2) = \sin^2(-0.000158) = 0.0000000249$$

$$\cos(\text{lat}_{\text{Malang}}) = \cos(-0.138659) = 0.990402$$
$$\cos(\text{lat}_{\text{Batu}}) = \cos(-0.138975) = 0.990359$$
$$\cos(\text{lat}_{\text{Malang}}) \times \cos(\text{lat}_{\text{Batu}}) = 0.980854$$

$$\sin^2(\Delta\text{lon}/2) = \sin^2(0.000163) = 0.0000000266$$

$$a = 0.0000000249 + 0.980854 \times 0.0000000266$$
$$a = 0.0000000511$$

#### Step 4: Hitung 'c' (Central Angle)

$$c = 2 \times \text{atan2}(\sqrt{a}, \sqrt{1-a})$$

$$\sqrt{a} = \sqrt{0.0000000511} = 0.0002259856$$

$$\sqrt{1-a} = \sqrt{0.9999999489} = 0.9999999745$$

$$\text{atan2}(0.0002259856, 0.9999999745) = 0.0002259856$$

$$c = 2 \times 0.0002259856 = 0.0004519711 \text{ radian}$$

#### Step 5: Hitung Jarak (d)

$$d = R \times c$$

$$d = 6371 \text{ km} \times 0.0004519711$$

$$d = 2.88 \text{ km}$$

**HASIL AKHIR**

**Jarak antara Hotel dan Wisata = 2.88 km**

## 4. IMPLEMENTASI DI EXCEL

Untuk mengimplementasikan formula Haversine di Microsoft Excel, gunakan formula berikut:

```
=6371 * 2 * ASIN(SQRT(
    POWER(SIN(RADIANS(Lat2-Lat1)/2), 2) +
    COS(RADIANS(Lat1)) * COS(RADIANS(Lat2)) *
    POWER(SIN(RADIANS(Lon2-Lon1)/2), 2)
))
```

### Contoh Penggunaan:

Jika koordinat Hotel ada di cell C41 (Lat) dan C42 (Lon), serta koordinat Wisata di C48 (Lat) dan C49 (Lon), maka formulanya:

```
=6371 * 2 * ASIN(SQRT(
    POWER(SIN(RADIANS(C48-C41)/2), 2) +
    COS(RADIANS(C41)) * COS(RADIANS(C48)) *
    POWER(SIN(RADIANS(C49-C42)/2), 2)
))
```

## 5. KEUNGGULAN FORMULA HAVERSINE

| No | Keunggulan                | Keterangan                                           |
|----|---------------------------|------------------------------------------------------|
| 1  | Akurat untuk jarak pendek | Error < 0.5% untuk jarak < 500 km                    |
| 2  | Simpel dan mudah          | Dapat diimplementasi di Excel tanpa library tambahan |
| 3  | Standard industri         | Digunakan di banyak aplikasi GPS dan navigasi        |
| 4  | Tidak butuh internet      | Perhitungan offline, tidak ada kuota/limit           |
| 5  | Gratis                    | Tidak ada biaya API atau licensing                   |
| 6  | Reproducible              | Dapat diverifikasi dan diaudit dengan mudah          |

## 6. REFERENSI

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Wikipedia. "Haversine formula". [https://en.wikipedia.org/wiki/Haversine\\_formula](https://en.wikipedia.org/wiki/Haversine_formula)

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